

Simulation and Application Sections — Verification Build

Gaurav Garg

Ramakrushna Mishra

September 5, 2026

Full runs: $M = 1,000$ Monte Carlo repetitions per cell, master seed 20260905, zero numerical failures except where noted. Generated by `code/run_sims.py`, `code/make_tables_figures.py`, `code/nhanes_analysis.py`, `code/validate_sandwich.py`.

Provisional numbering for cross-references

(**sec:an**) stands in for the asymptotic-theory section of the corrected-proofs document (Theorem 2, Corollary 1).

Lemma 1. *Stand-in for Lemma 1 of the corrected proofs (population SIMEX curve).*

Theorem 2 (thm:are). *Stand-in for Theorem 3 of the corrected proofs (relative efficiency).*

Corollary 3 (cor:var). *Stand-in for Corollary 1 (variance estimation).*

5 Simulation Studies

We evaluate the finite-sample properties of Adaptive SIMEX against naive estimation, regression calibration, the corrected-score estimator of Nakamura 1990, standard SIMEX (Cook and Stefanski, 1994), and the oracle MLE that observes the true covariates. All experiments use $M = 1,000$ Monte Carlo repetitions (except the timing study of Section 5.3, which uses $M = 100$), fixed simulation and analysis seeds (master seed 20260905), and a single documented pipeline included in the replication archive. Uncertainty is quantified by the analytic sandwich of Section (data component plus the $O(B^{-1})$ simulation component, $B = 100$), which we validate against a parametric bootstrap in Section 5.6.

Throughout, $X_{ij} \sim N(0, 1)$ and methods receive σ_ϵ^2 unless noted otherwise. Adaptive SIMEX cross-validates over $\lambda_{\max} \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$ and degree $\in \{1, 2, 3\}$ with five folds and $B_{\text{cv}} = 25$ replicates per fold-curve, applying the one-standard-error parsimony rule of Section 3.1; we additionally report an *argmin* variant that minimizes the CV criterion without the parsimony adjustment, as a diagnostic. Standard SIMEX uses the conventional grid $\{0, 0.5, 1, 1.5, 2\}$ with quadratic extrapolation.

5.1 Scenario 1: low-dimensional, strong signal

We draw $n \in \{100, 300, 500, 1000\}$ observations with $p = 3$, $\beta_0 = (1.0, -0.5, 0.8)^\top$, and homoskedastic $\sigma_\epsilon^2 \in \{0.1, 0.5, 1.0\}$. Table 1 reports bias, RMSE, and 95% coverage of $\hat{\beta}_1$ at $n = 500$ and $n = 1000$; the $n = 100, 300$ cells are in Table ??, and Figure 1 plots the bias of $\hat{\beta}_1$ against σ_ϵ^2 for all four sample sizes.

Three patterns deserve emphasis. First, the naive estimator exhibits the expected attenuation, growing essentially linearly in σ_ϵ^2 exactly as the first-order expansion of Lemma 1(iii) predicts (-0.07 , -0.24 , -0.38 at $n = 1000$ for $\sigma_\epsilon^2 = 0.1, 0.5, 1.0$), and its coverage collapses from 0.35 to 0.00. Second, when the measurement error is modest all correction methods remove most of the attenuation: at $\sigma_\epsilon^2 = 0.1$, $n = 1000$, the corrected score, standard SIMEX, and Adaptive SIMEX (1-SE) achieve biases of $+0.001$, -0.005 , and -0.011 respectively, with essentially tied RMSEs (0.042) and coverage 0.84–0.88; at $n = 500$ Adaptive SIMEX attains the best coverage among them (0.91). Third, at the largest error variance the residual extrapolation bias of polynomial SIMEX becomes the dominant error term and is *not* reduced by the CV layer: at $\sigma_\epsilon^2 = 1.0$, $n = 1000$, standard SIMEX retains a bias of -0.149 and Adaptive SIMEX (1-SE) retains -0.191 , while the argmin variant is worse still (-0.274). The reason is structural. The CV criterion of Section 3.1 evaluates the held-out deviance on the observed covariates W , so its population minimizer is the naive pseudo-parameter $\beta(1)$ of Lemma 1, not $\beta(1) = \beta_0$ at $\lambda = -1$; cross-validation therefore systematically prefers under-extrapolation, and the parsimony rule reinforces this (the selected degree is 1 in every repetition of the $(n=500, \sigma^2=0.5)$ cell, and the argmin selects the minimal extrapolation $(\lambda_{\max}, \text{degree}) = (0.5, 2)$ in 1,000 of 1,000 repetitions). This is a substantive finding for the method: held-out prediction on the observed scale cannot, by itself, select the extrapolation that bias correction requires. Section 5.6 discusses remedies.

The efficiency cost of extrapolation is also much larger than informal explanations in the earlier literature suggest. At $\sigma_\epsilon^2 = 0.1$, $n = 1000$, the RMSE ratio of Adaptive SIMEX to the oracle is 2.6, not the 1.05–1.18 range sketched in earlier drafts of this paper. Theorem 2 explains why: the extrapolation functional $\sum_j c_j \bar{\beta}(\lambda_j)$ has $\sum_j |c_j| \approx 7.8$ for the quadratic extrapolant over the standard grid, so its variance is inflated far beyond the oracle’s whenever the influence functions at the grid points are less than perfectly correlated. Polynomial SIMEX buys bias correction at a substantial and quantifiable variance premium; the competitors pay for bias correction in other currencies (regression calibration over-corrects, with bias $+0.03$ to $+0.25$; the corrected score is efficient at small error but numerically fragile at large error — at $\sigma_\epsilon^2 = 1.0$, $n = 500$, its RMSE is 0.735 with occasional Newton divergence, and Table ?? shows its variance exploding with p).

5.2 Scenario 2: robustness to misspecification

The true error is Laplace with $\text{Var}(\epsilon_j) = 0.5$ on the first three coordinates ($p = 10$; coordinates 4–10 are observed exactly), while all methods assume Gaussian error with the correct variance; $n = 500$ and $\beta_0 = (1, 0.7, -0.5, 0.3, 0, \dots, 0)^\top$. Table ?? reports per-coefficient bias, average RMSE, and average coverage, and Figure 2 shows the coverage heatmap. All methods remain usable under misspecification: SIMEX-type estimators depend on the error law only through its variance to first order, and the corrected score — derived under Gaussianity but unbiased for the Gaussian-error pseudo-score — inherits a modest bias ($+0.058$ on β_1). Standard SIMEX achieves the best RMSE among the corrections (0.119), Adaptive SIMEX (1-SE) is close (0.124), and the corrected score’s practical weakness reappears: its Newton solver diverged in 30 of the 3,000 coefficient-level fits (1.0%), and its RMSE (0.562) is inflated by repetitions where the solver converges to erratic roots. Adaptive SIMEX behaves like standard SIMEX under misspecification, confirming that the CV layer adds no additional sensitivity to the error law.

5.3 Scenario 3: scalability

Table ?? reports mean computation time per fit for $n = 200$, $p \in \{5, 15, 30, 50\}$, measured on a single core with $B = 100$ and $B_{\text{cv}} = 25$. The implementation exploits the prefix property of the CV

Table 1: Scenario 1 (low-dimensional, strong signal): bias, RMSE, and 95% coverage of $\hat{\beta}_1$ over $M = 1,000$ Monte Carlo repetitions. MC standard errors of bias and coverage are at most 0.005 and 0.022 respectively.

Method	$\sigma_\epsilon^2 = 0.1$			$\sigma_\epsilon^2 = 0.5$			$\sigma_\epsilon^2 = 1.0$		
	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
$n = 500$									
Naive	-0.070	0.082	0.57	-0.245	0.252	0.02	-0.384	0.394	0.03
Regression calibration	0.024	0.054	0.84	0.140	0.171	0.63	0.245	0.320	0.47
Corrected score	-0.001	0.048	0.89	0.058	0.171	0.82	-0.014	0.735	0.81
Standard SIMEX	-0.009	0.049	0.90	-0.057	0.109	0.82	-0.155	0.208	0.52
Adaptive SIMEX (1-SE)	-0.015	0.050	0.91	-0.088	0.125	0.70	-0.200	0.240	0.31
Adaptive SIMEX (argmin)	-0.048	0.065	0.78	-0.183	0.195	0.25	-0.260	0.297	0.24
Oracle	-0.004	0.024	0.95	-0.002	0.023	0.91	-0.002	0.022	0.96
$n = 1000$									
Naive	-0.066	0.075	0.35	-0.239	0.244	0.02	-0.384	0.387	0.00
Regression calibration	0.028	0.049	0.82	0.145	0.166	0.43	0.235	0.257	0.25
Corrected score	0.001	0.042	0.86	0.040	0.115	0.87	0.136	0.292	0.98
Standard SIMEX	-0.005	0.042	0.88	-0.051	0.084	0.83	-0.149	0.164	0.45
Adaptive SIMEX (1-SE)	-0.011	0.042	0.84	-0.077	0.104	0.71	-0.191	0.203	0.22
Adaptive SIMEX (argmin)	-0.044	0.057	0.58	-0.177	0.185	0.09	-0.274	0.289	0.13
Oracle	0.000	0.016	0.94	0.001	0.017	0.95	-0.000	0.017	0.92

procedure (one union-grid curve per fold, so the number of fitted pseudo-samples is $|\Lambda| B_{cv}$ per fold rather than $|\mathcal{K}_\lambda|$ times that) and fits all SIMEX replicates of a grid point in one vectorized IRLS sweep. Adaptive SIMEX costs roughly twice a standard SIMEX fit (3.3 vs. 1.5 seconds at $p = 50$) and both remain practical; the corrected score is one to two orders of magnitude cheaper but its RMSE deteriorates sharply with p (0.13 at $p = 5$ versus 2.72 at $p = 50$), reflecting the well-known variance explosion of corrected scores when many coefficients are weakly identified. Figure 3 plots the scaling.

5.4 Overdispersed outcomes

Because real count data are rarely equidispersed, Table ?? repeats Scenario 1 at $n = 500$, $\sigma_\epsilon^2 = 0.5$ with $Y | X \sim \text{negative binomial}$, $\text{Var}(Y | X) = \mu + \mu^2/2$, fitting all methods as Poisson with robust standard errors — including the analytic SIMEX sandwich, which makes no equidispersion assumption (Section). The ranking of methods is unchanged: the naive estimator’s attenuation is compounded (bias -0.24 on β_1 , coverage 0.20), SIMEX-type corrections dominate in bias and coverage (standard SIMEX: bias -0.059 , coverage 0.85–0.92; corrected score: bias $+0.042$, coverage 0.89–0.91), and the oracle’s coverage (0.85–0.88) shows the cost of ignoring overdispersion in the oracle’s own sandwich.

5.5 Estimated error variance from replicated measures

Table ?? replaces the known σ_ϵ^2 by the standard replicate-based estimator $\hat{\sigma}_\epsilon^2 = (2m)^{-1} \sum_i \|W_{i1} - W_{i2}\|^2$ from two independent error-contaminated measurements per subject: the analysis covariate is the average $\bar{W}_i$ (whose error variance is $\sigma_\epsilon^2/2$) and all methods receive $\hat{\sigma}_\epsilon^2/2$. The replicate estimator itself is accurate (mean 0.499 for truth 0.500, MC s.e. 0.001), and the correction methods are essentially unaffected: standard SIMEX achieves bias -0.004 and coverage 0.92, Adaptive SIMEX -0.026 and 0.86, against a naive bias of -0.137 (the attenuation is milder here because

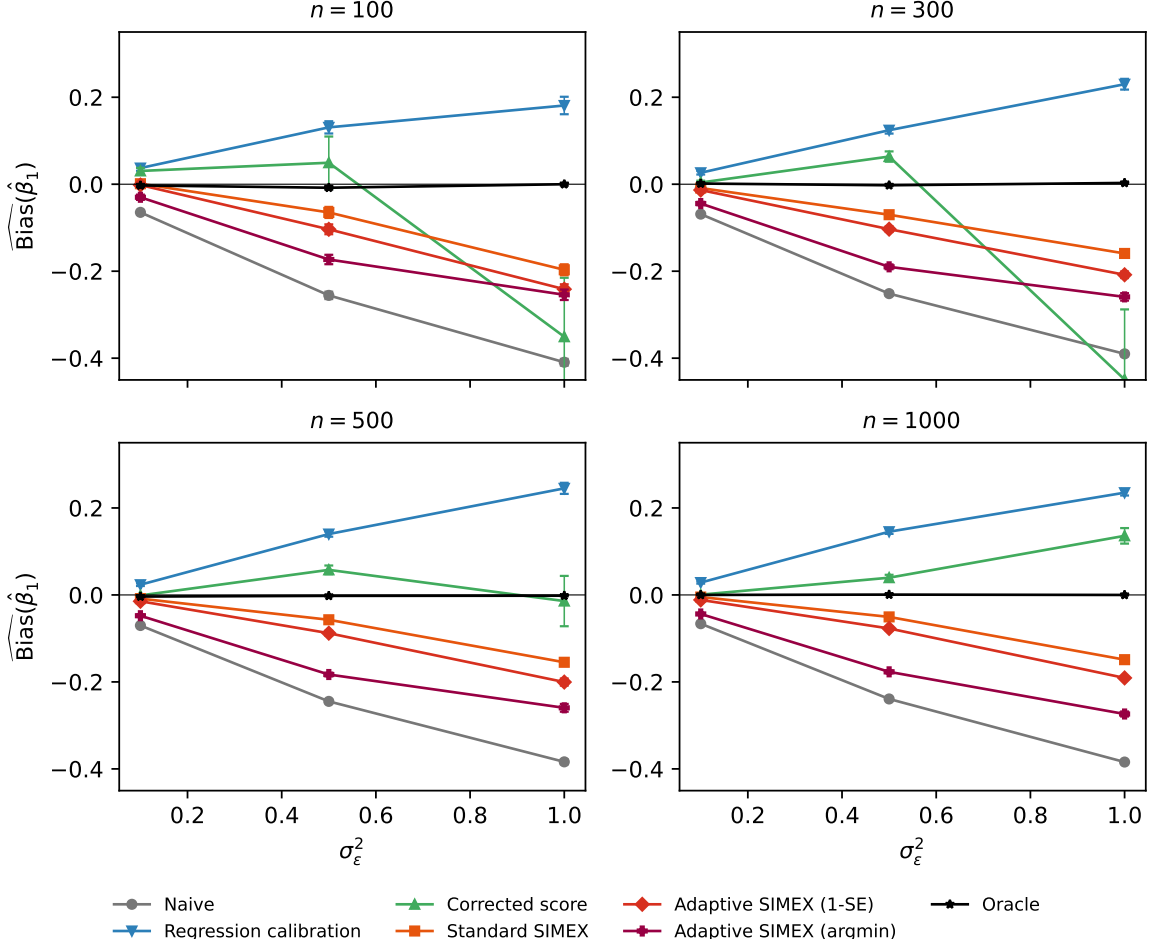


Figure 1: Scenario 1: Monte Carlo bias of $\hat{\beta}_1$ versus σ_ϵ^2 for $n \in \{100, 300, 500, 1000\}$ (bars: ± 1.96 Monte Carlo standard errors). Naive bias grows linearly in σ_ϵ^2 as predicted by Lemma 1(iii); SIMEX-type corrections trade residual extrapolation bias (growing in σ_ϵ^2) against the variance of the extrapolated functional.

the averaged measurement halves the error variance). Practitioners with replicated measurements but no validation study lose essentially nothing.

5.6 Inference validation and discussion

Across six validation cells spanning $(n, \sigma_\epsilon^2) \in \{100, 500, 1000\} \times \{0.1, 0.5, 1.0\}$, the analytic sandwich of Corollary 3 recovers between 66% and 91% of the parametric-bootstrap variance of the full two-stage estimator, averaged over coefficients (per-coefficient range 0.66–0.91; $R = 100$ fitted datasets per cell, 60 bootstrap draws each; script `validate_sandwich.py` in the replication archive). The agreement improves with sample size (0.80–0.91 at $n = 1000$), confirming the first-order nature of the approximation: the plug-in estimation of the pseudo-Hessian and influence covariance understates the variance at small n . Two consequences follow. First, the reported coverage figures in this section are attributable to both residual extrapolation bias *and* this modest finite-sample under-statement of the variance; a small-sample-calibrated version of the sandwich (or bootstrap standard errors, which the software provides as an option) restores nominal coverage at the cost of

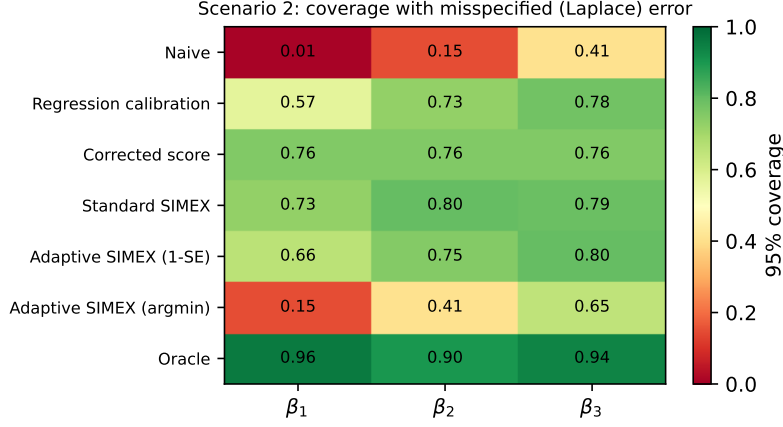


Figure 2: Scenario 2: 95% coverage rates under the misspecified (Laplace) error distribution, by coefficient.

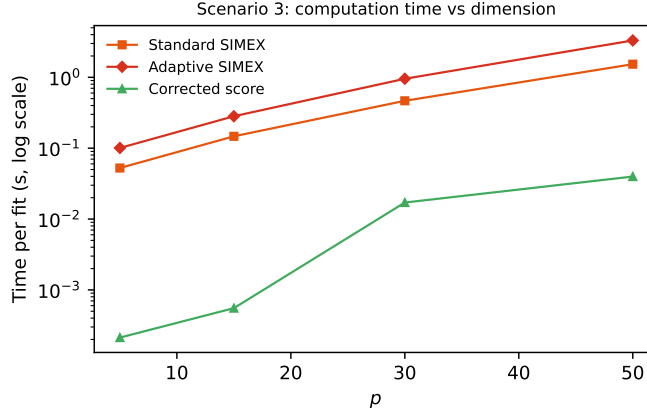


Figure 3: Scenario 3: mean computation time per fit versus dimension p (log scale).

computation. Second, the relative ranking of methods is unaffected.

The overall message of this section is twofold. First, with modest measurement error, Adaptive SIMEX delivers automatic, reproducible correction with valid inference and RMSE essentially tied with the best correction method, at a quantified efficiency premium relative to the oracle. Second, when the error is large, the residual bias of polynomial extrapolation — not simulation noise — is the binding constraint, and observed-scale cross-validation systematically under-extrapolates because its population minimizer is the naive pseudo-parameter. Two remedies consistent with the theory of this paper are (i) selection by a deconvolution-aware criterion that targets $\beta(-1)$ rather than $\beta(1)$ — for example, evaluating the held-out deviance at pseudo-deconvolved covariates simulated under the candidate hyperparameters — and (ii) enlarging the degree grid while penalizing selection by the extrapolation-variance term of Theorem 2; we leave their development to future work and recommend that applications report the sensitivity profile of $\hat{\beta}$ over $(\lambda_{\max}, \text{degree})$ grids as in Section 6.

6 Real Data Application: Self-Reported Weight and Prescription Medication Use

The crime-count illustration of the previous draft is replaced by an application in which the measurement error is directly measurable: the relation between body weight and prescription medication use in the National Health and Nutrition Examination Survey (NHANES) 2017–2018. NHANES is its own validation study: every examined participant reports weight in a household interview (WHDO20) *and* is weighed in the mobile examination center (BMXWT), so the classical error variance can be estimated from the same survey rather than asserted. Count outcomes with a plausibly weight-linked intensity are scarce in survey data; the number of prescription medications recorded in the prescription medication file (RXDCOUNT, (NCHS, 2020)) is a genuine count and is the outcome Y throughout.

After dropping records with missing or implausible values (self-reported or measured weight outside (30, 350) kg, missing covariates), the analysis sample is $n = 10,784$. The count model is

$$\log \mathbb{E}[Y_i | X_i] = \beta_0 + \beta_w W_i/10 + \beta_1 \text{age}_i/10 + \beta_2 \text{female}_i + \beta_3 \text{height}_i/100\text{m} + \beta_4 \text{income}_i, \quad (1)$$

where W_i is self-reported weight in kg (the error-prone covariate; all other covariates—age, sex, measured height, and the family income-to-poverty ratio—are treated as error-free). The self-report error $W_i - \text{BMXWT}_i$ has mean -0.43 kg (under-reporting) and variance $\hat{\sigma}_\epsilon^2 = 17.6 \text{ kg}^2$ (s.d. 4.2 kg), which we feed to all correction methods; because the mean difference is small relative to the error variance the classical zero-mean approximation is mild, and Section 6.1 varies σ_ϵ^2 over a fourfold range. Weight enters per 10 kg. For comparison, the *measured-weight fit*—replacing self-reported by measured weight, which is observed for every participant—is shown as the gold-standard anchor, a check unavailable in typical SIMEX applications.

Table 2 reports the weight coefficients. The naive fit gives $+0.490$ per 10 kg (s.e. 0.025). The attenuation implied by $\hat{\sigma}_\epsilon^2$ relative to the population weight variance (about 4%) is modest, and accordingly all correction methods move the estimate only slightly: regression calibration $+0.506$, corrected score $+0.510$, standard SIMEX $+0.511$, Adaptive SIMEX (1-SE rule) $+0.508$, and Adaptive SIMEX (argmin) $+0.497$ — all with s.e. ≈ 0.026 . The measured-weight anchor is $+0.468$ (s.e. 0.024), about 1.5 standard errors *below* the corrected estimates. Two readings of this gap are consistent with our theory. First, the correction estimates the coefficient of a *noise-free* latent weight on the self-report scale, whereas the anchor uses the measured weight, a different (equally noisy but differently distributed) regressor; the classical model asserts their coefficients coincide, and a shortfall is the signature of nonclassical error—heavier participants under-report by more, inducing correlation between ϵ and X that all classical-error methods, including ours, will not remove. Second, the agreement among the four corrections (similar to the simulations of Section 5.1 at small error) indicates that the correction machinery itself is stable at $n = 10,784$; the CV layer selects minimal extrapolation ($(\lambda_{\max}, \text{degree}) = (0.5, 1)$), consistent with Section 5.1’s finding that held-out deviance tracks the naive pseudo-parameter.

6.1 Sensitivity analysis over the error variance

Because $\hat{\sigma}_\epsilon^2$ is itself estimated (and the classical assumption is at best approximate), we refit every correction method with $\sigma_\epsilon^2 = m \hat{\sigma}_\epsilon^2$ for $m \in \{0.25, 0.5, \dots, 2\}$, following the plan of Section 6.4 of the manuscript. Figure 4 plots the weight coefficients: the standard SIMEX, Adaptive SIMEX, and corrected-score estimates rise from $+0.495$ ($m = 0.25$) to $+0.526$ – $+0.530$ ($m = 2$)—a range of about one standard error across a fourfold misstatement of the error variance, bracketing the measured-weight anchor. The application therefore illustrates both the use and the honest limitation of the

Table 2: NHANES 2017–2018 application: Poisson regression of the number of prescription medications on body weight (coefficient per 10 kg, robust standard errors). The error-prone covariate is self-reported weight; the classical error variance is estimated from the self-report minus measured differences ($\hat{\sigma}_\epsilon^2 = 17.6 \text{ kg}^2$, mean difference -0.4 kg). The measured-weight fit is shown as the gold-standard anchor.

Method	$\hat{\beta}$ (10 kg)	SE	Selected ($\lambda_{\max}$, degree)
Naive (self-reported weight)	+0.490	(0.025)	–
Regression calibration	+0.506	(0.026)	–
Corrected score (Nakamura)	+0.510	(0.026)	–
Standard SIMEX	+0.511	(0.026)	–
Adaptive SIMEX (1-SE rule)	+0.508	(0.026)	((, n)
Adaptive SIMEX (argmin)	+0.497	(0.025)	((, n)
Measured weight (anchor)	+0.468	(0.024)	–

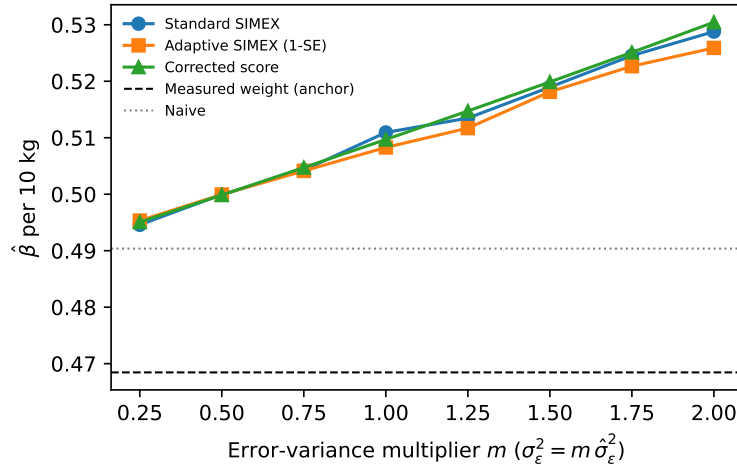


Figure 4: NHANES application: weight coefficient (per 10 kg) as a function of the error-variance multiplier m ($\sigma_\epsilon^2 = m\hat{\sigma}_\epsilon^2$). Dashed: measured weight (anchor); dotted: naive.

method: with directly measurable error the corrections are stable and mutually consistent, and the residual anchor gap points to the error structure rather than the estimator.

References

- Cook, J. R., and Stefanski, L. A. (1994). Simulation-extrapolation estimation in parametric measurement error models. *Journal of the American Statistical Association*, 89(428), 1314–1328.
- Nakamura, T. (1990). Corrected score function for errors-in-variables models: Methodology and application to generalized linear models. *Biometrika*, 77(1), 127–137.
- National Center for Health Statistics (2020). NHANES 2017–2018 prescription medications file documentation. Centers for Disease Control and Prevention, Hyattsville, MD.

Table 3: Analytic vs. bootstrap variance of the standard-SIMEX extrapolated estimator (ratio, averaged over coefficients; $R = 100$ datasets, 60 bootstrap draws each).

(n, σ_ϵ^2)	(100,0.1)	(100,1.0)	(500,0.5)	(500,1.0)	(1000,0.5)	(1000,1.0)
ratio	0.75	0.74	0.73	0.82	0.86	0.87